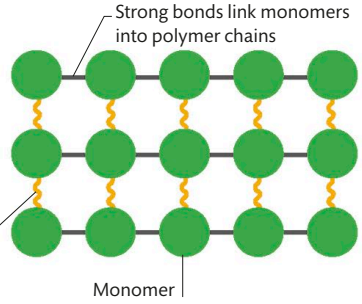
Plastics

Plastics are synthetic materials made of polymers—long-chain molecules consisting of repeating units called monomers. Due to their low cost, ease of manufacture, and versatility, plastics are one of the most widely used types of materials in the modern world.

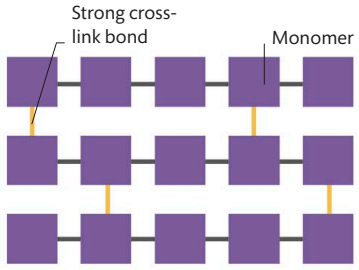
Types of plastic

There are two main types of plastic. Thermoplastics are easy to melt and recycle. Examples include polyethylene, polystyrene, and polyvinyl chloride (PVC). Thermosetting plastics are hardened by heat and cannot be remelted. Less commonly used than thermoplastics, thermosetting plastics include polyurethane, melamine, and epoxy resins.

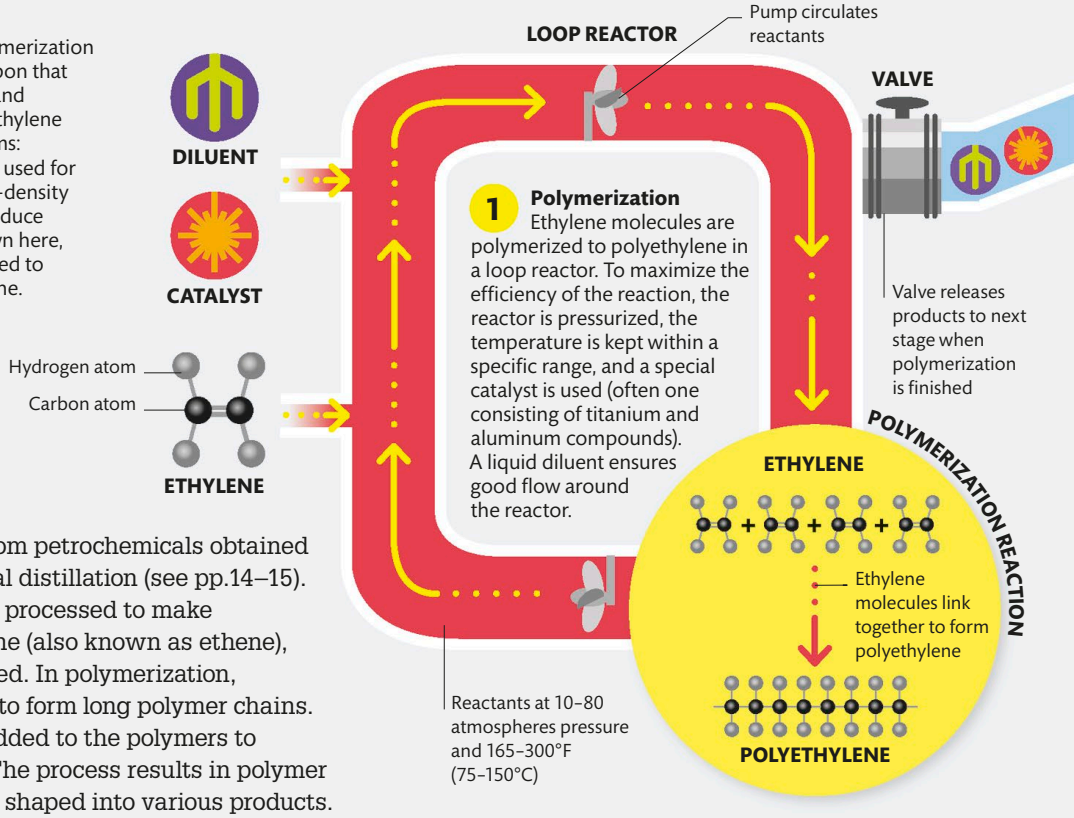
Thermoplastics
In a thermoplastic, long polymer molecules are joined to one another by weak bonds that easily break apart when the plastic is heated and quickly reform again as it cools.



Thermosetting plastics
Thermosetting plastics have strong cross-link bonds binding the polymer chains. The plastics are soft at low temperatures and then permanently set (hardened) by the application of heat.



Making polyethylene
Polyethylene is made by the polymerization of ethylene, a colorless hydrocarbon that is gaseous at room temperature and which is obtained from oil. Polyethylene is manufactured in two main forms: low-density polyethylene (LDPE), used for plastic bags and sheets, and high-density polyethylene (HDPE), used to produce harder plastics. The process shown here, known as the slurry process, is used to produce high-density polyethylene.



Making plastics

Most plastics are made from petrochemicals obtained from crude oil by fractional distillation (see pp.14–15). These petrochemicals are processed to make monomers such as ethylene (also known as ethene), which are then polymerized. In polymerization, monomers react together to form long polymer chains. Other chemicals can be added to the polymers to change their properties. The process results in polymer resins, which can then be shaped into various products.